



King Faisal University

Collage of Computer Science and Information Technology

Fundamental Of Software Engineering

Hotel Reception System

Group:GF21

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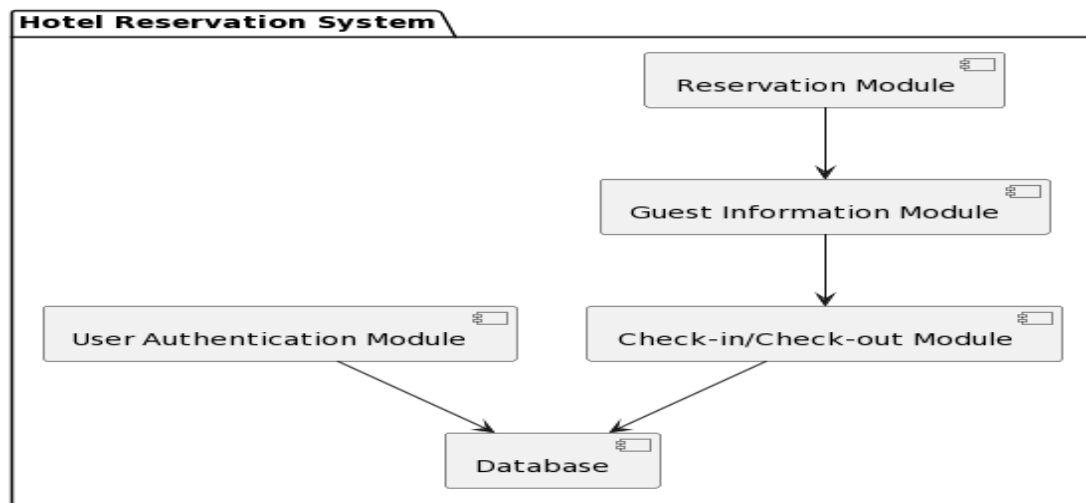
Introduction:

The hotel reservation and guests management system aim to streamline the booking process and enhance overall guest experience. Efficiently manage reservations, check-ins, and check-outs. Ensure accurate guest information tracking. The system will cover reservation management, guest information storage, check-in/check-out processes, and room availability tracking.

User requirement Definition:

- 1.1 The system shall verify employee number to perform login.
- 1.2 The system shall allow the employee to add a new customer by filling a form of his information to complete the check in process.
- 1.3 The system shall display the room list.
- 1.4 The system shall display customer information.
- 1.5 The system shall do the check out by providing the customer id.

System Architecture:



System Requirements Specification:

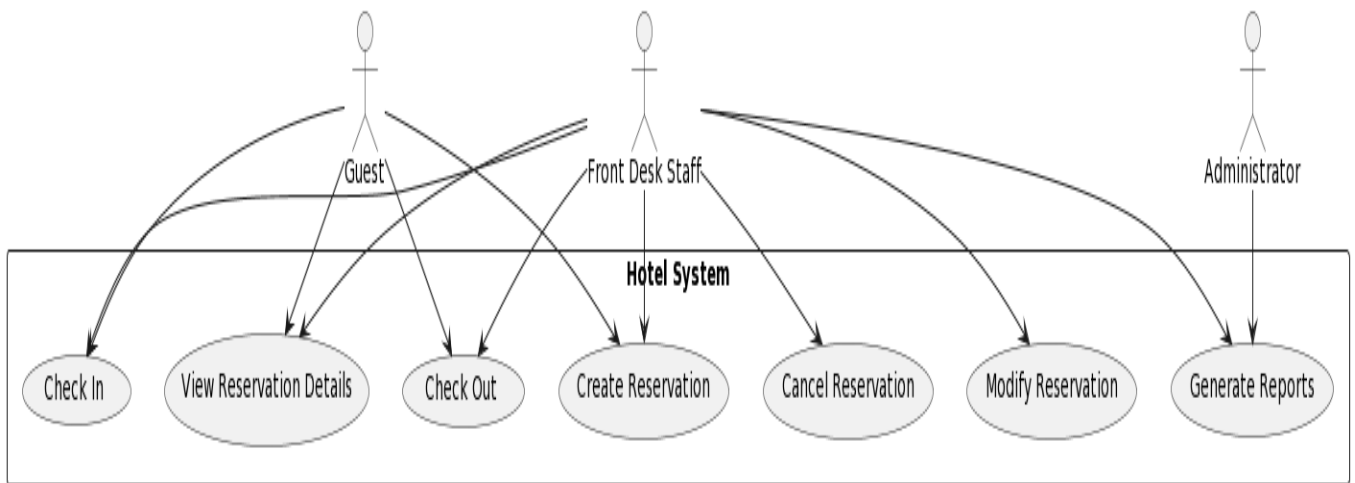
Functional Requirements:

1. Ability to create, modify, and cancel reservations.
2. Real-time room availability updates.
3. Guest check-in and check-out processes.
4. Secure user authentication.
5. Reporting and analytics functionality.

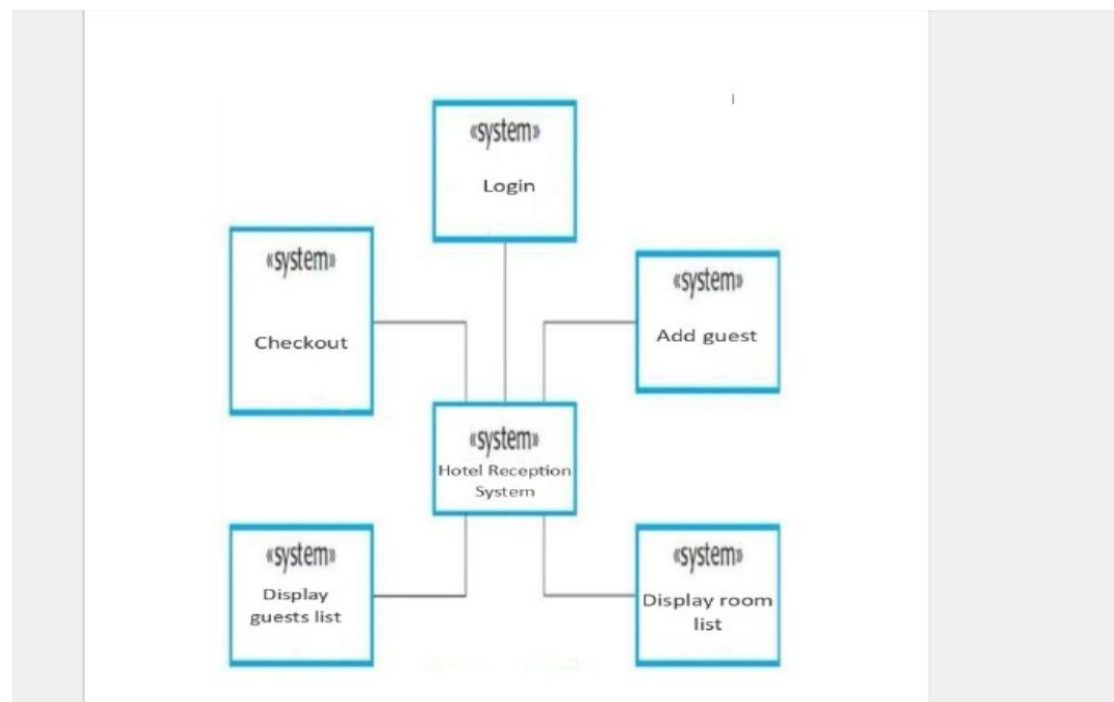
Non-functional Requirements:

1. System should be available 24/7.
2. Response time for operations should be within 2 seconds.
3. Data backup and recovery mechanisms.
4. System should support at least 1000 concurrent users.

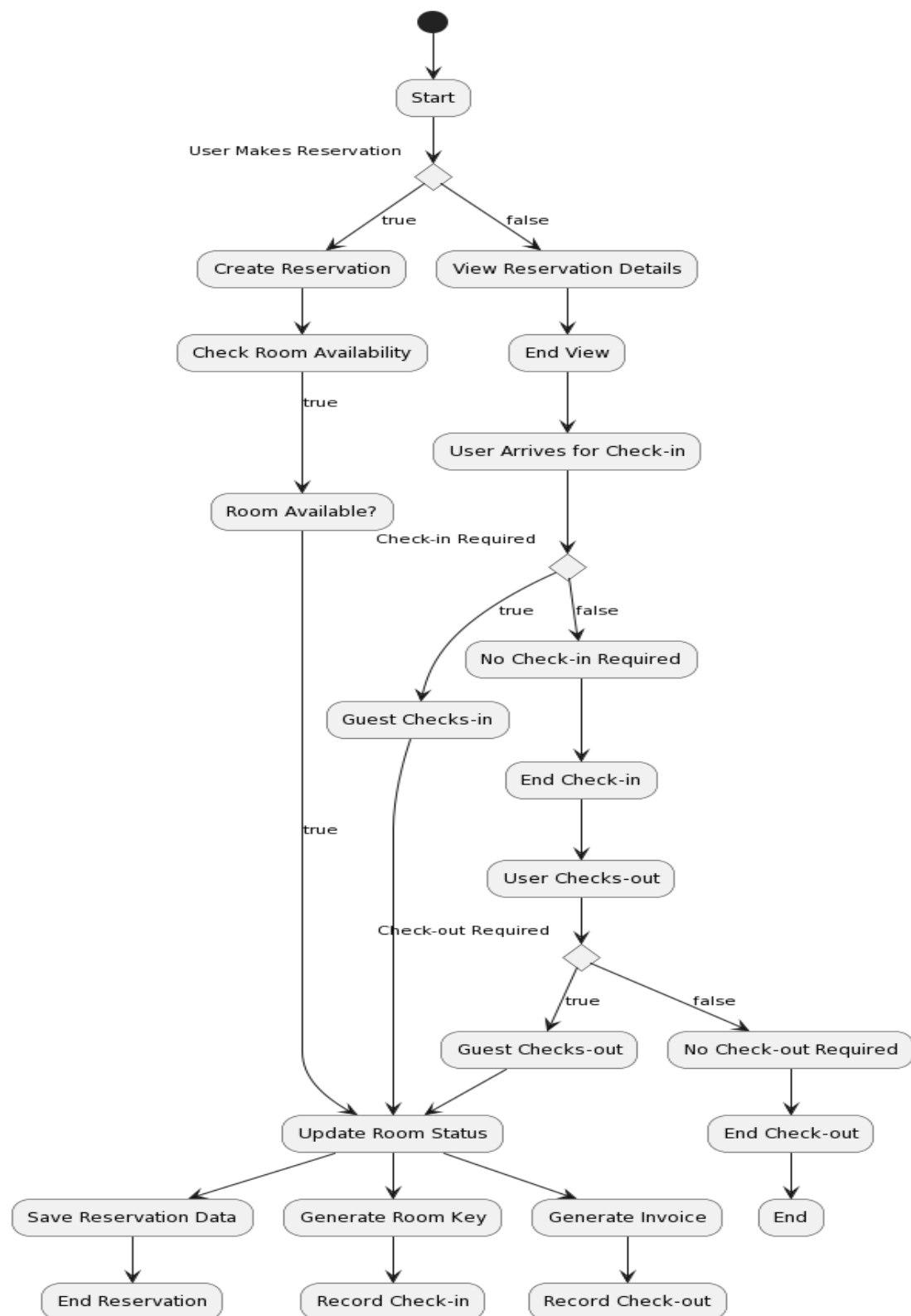
Use case Diagram:



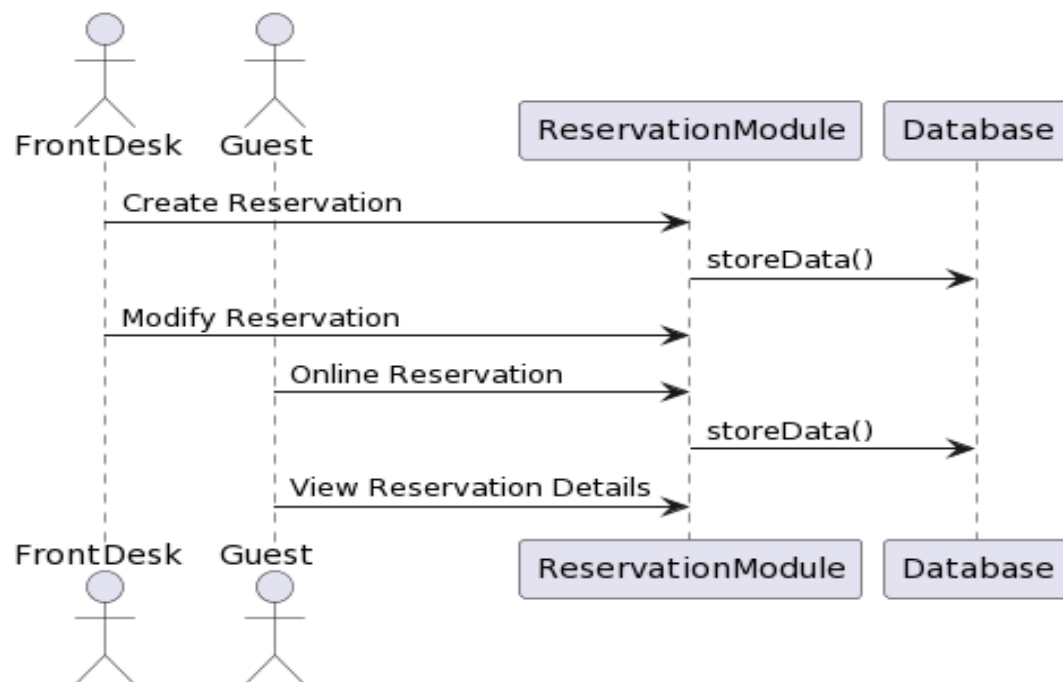
Context model:



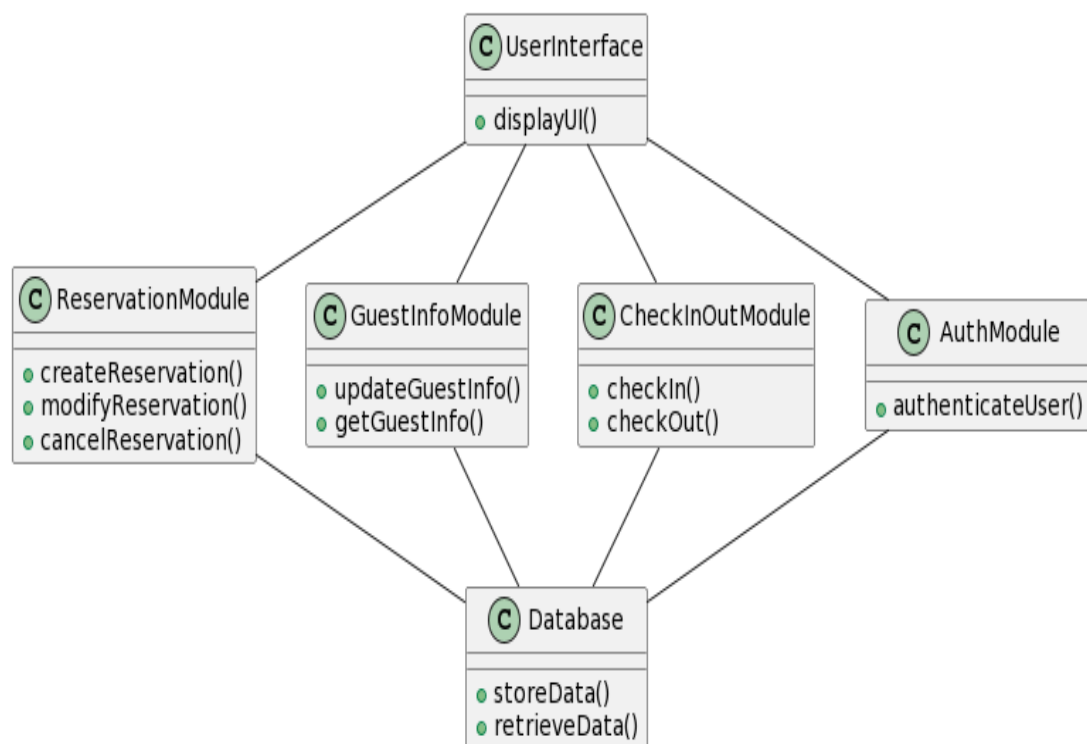
Process model:



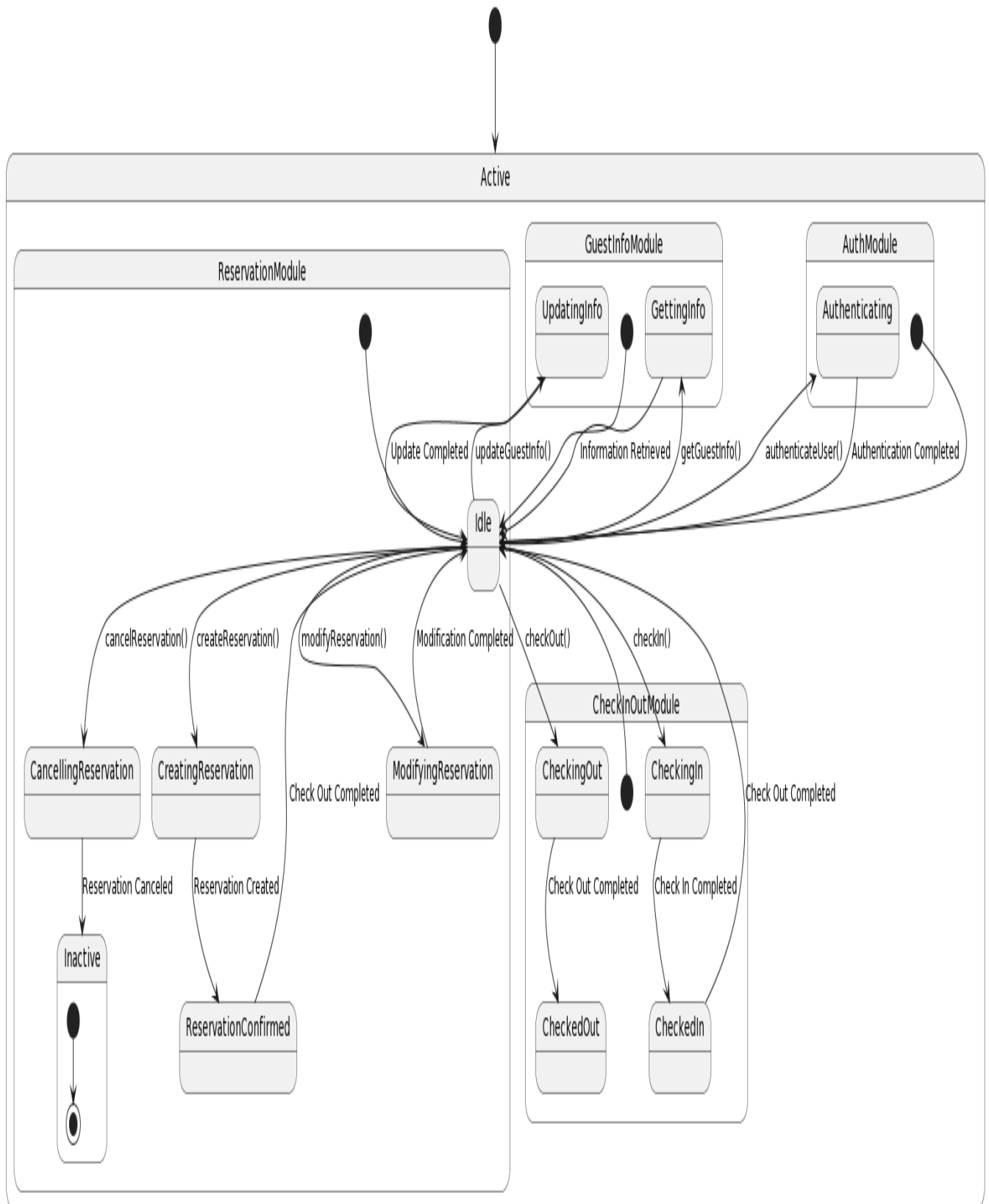
Sequence diagram:



Class diagram:



State Machine diagram:



Test cases:**1.Login:**

Test case 1: Verify that a valid username and password combination allows the receptionist to log in successfully.

Test case 2: Verify that an invalid username and password combination does not allow the receptionist to log in.

2.Add guest:

Test case 1: Verify that all required guest information (such as name and ID) is correctly entered and saved.

Test case 2: Verify that the system does not allow adding a guest without providing information.

3.Display rooms list:

Test case 1: Verify that all available rooms are displayed correctly, including their room numbers, ID, and occupancy.

Test case 2: Verify that the system correctly updates the availability status of rooms when they are booked or checked out.

4.Display guests list:

Test case 1: Verify that all guests currently staying at the hotel are - displayed correctly, including their names, room numbers, and check-in/check-out dates

Test case 2: Verify that the system correctly updates the guest list - when guests check in or check out.

5.checkout:

Test case 1: Verify that the system correctly checkout the guest.

Test case 2: Verify that the system marks the room as available after the guest checks out.

Task, Duration, and Dependencies table:

Task ID	Task Description	Duration (weeks)	Dependencies
T1	Project Planning	2	-
T2	Requirements Analysis	3	T1
T3	System Design	4	T2
T4	Database Setup	2	T3
T5	Frontend Development	4	T3
T6	Backend Development	5	T3
T7	Integration Testing	3	T4, T5, T6
T8	User Acceptance Testing (UAT)	2	T7
T9	Documentation	2	T2, T6
T10	Deployment	2	T8, T9

Activity Bar Charts:

Task	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Define Requirements	**	----- -----				
System Design		**	----- -----			
Database Design		**	----- -----			
Module Development			**	----- -----		
UI Development				**	----- -----	
Integration Testing				**	----- -----	
User Acceptance Testing					**	----- -----
Documentation					**	----- -----
Deployment					**	-----

Staff Allocation Charts:

Task	Developer 1	Developer 2	UI Designer	Tester	Documenter
Define Requirements	**				
System Design	**	**			
Database Design		**			
Module Development	**	**			
UI Development			**		
Integration Testing	**	**	**	**	
UAT				**	
Documentation	**	**	**	**	**
Deployment	**	**	**	**	**